

3

Improvement in Food Resources



“Food is the combination of various organic and inorganic substances which are capable of providing energy for the various metabolic activities.”

Introduction

Food is any substance or material consumed to provide nutritional support for the body. It is usually of plant or animal origin. Food contains essential nutrients, such as carbohydrates, fats, proteins, vitamins, minerals which is ingested and assimilated by an organism to produce energy, stimulate growth and maintain life.

Food Sources

- (i) **Plants:** Many plants or plant parts are eaten as food. Plant provides cereals, vegetables, spices and fruits etc.
- (ii) **Animals:** Animals are used as food either directly or indirectly by the products they produce such as meat, milk, honey, eggs etc.

Why do we need to improve crops?

As the population of the world continues to increase, the demand for food also increases. Increased crop yields, pest resistance and tolerance to environmental stresses have been the targets of plant breeders. There is also progress towards improving food quality, as well as quantity, by improving the nutritional benefits of food crops.



Blue revolution - to increase fish production
Yellow revolution - to increase oil production
White revolution - to increase milk production
Silver revolution - to increase egg production

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Green Revolution and Its Impacts: The introduction of high-yielding varieties of seeds after 1965 & increased use of fertilizers & irrigation are known collectively as green revolution. It helped in increasing the production needed to make India self - sufficient in food grains. Father of green revolution in India is M.S. Swaminathan. The Green revolution brought about a great increase in the production of food grains especially wheat crop. Drawbacks of green revolution are the use of synthetic chemical pesticides and insecticides on a large scale, causing unimaginable damage to the fertility of the soil and destroying the micronutrients in it.

The hybrid variety of crops intended to boost productivity could not resist pests. Further only a few varieties of seed were promoted. This led to the loss of variety in seeds and crops. The excessive use of water that the green revolution promoted has led to the drying up of water sources.

Thus, it has become far more important that we should increase food production without degrading our environment and disturbing the natural balance. Therefore, there is a need of practising farming using principles of ecology. The practice of farming and production of maximum agriculture yield through management of natural resources without disturbing the environment is known as sustainable agriculture. It has the following objectives –

- (i) Satisfy human food and fiber needs.
- (ii) Make the most efficient use of nonrenewable resources and on-farm resources, natural biological cycles and controls.
- (iii) Farming methodology and equipments should be economical for the farmer.
- (iv) Enhance the quality of life for farmers and society as a whole.

Improvement in crop yields



Norman Borlaug Father of Green Revolution {World}

Crops: Plants of same kind which are grown and cultivated at one place on a large scale are known as crops.

Types of crops:

- (i) **Cereals:** These are rich source of carbohydrate and thus provide energy.
Example: Rice, wheat, maize, sorghum, barley, millets.
- (ii) **Pulses:** These are rich source of protein and thus helpful in body building.
Example: Gram(chana), moong (green gram), pea(matar), lentil(masoor), urad(blackgram), pigeon pea (arhar).



White revolution has led to better and more efficient use as well availability of milk.

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Norman E Borlaug was an American agronomist led initiatives worldwide that contributed to extensive increase in agricultural production termed the green revolution.

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(iii) **Oil seed crops:** These are rich source of oil and fatty acids.

They provide us necessary fats.

Example: Groundnut, linseed, sunflower, soyabean, mustard, sesame, castor etc.

(iv) **Vegetable crops:** These provide vitamins, minerals along with small quantities of carbohydrate, protein and fats.

(v) **Spices:** They are used for enhancing palatability.

Example: Chilly, black pepper, ginger, turmeric etc.

(vi) **Fodder crops:** These serve as green fodder to the cattle.

Example: Berseem, sorghum, maize, oat, sudan grass, alfalfa etc.

Different types of crops require different climatic conditions like temperature, humidity and photoperiods (duration of sunlight). Crops are divided into two groups on the basis of growing season.

(i) **Kharif crops:** These crops are sown in the months of June/July and harvested in September/October every year.

Example: Paddy, maize etc.

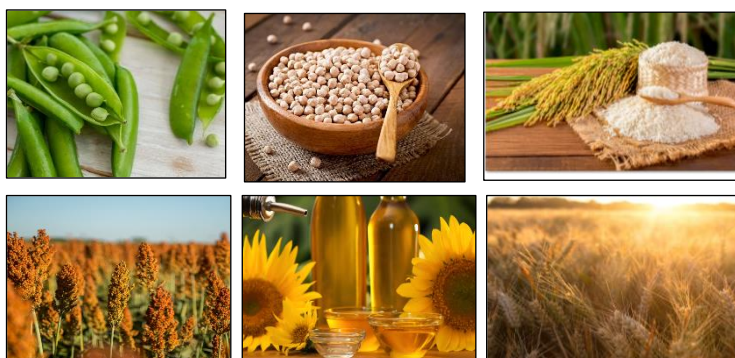
(ii) **Rabi crops:** These crops are sown in the months of October/November and harvested in March/April every year.

Example: Wheat, oat, barley and pea etc.



Zaid crops are grown on irrigated lands in the short duration between Rabi and Kharif crop season mainly from March to June e.g. Seasonal fruits and vegetables like watermelon, cucumber.

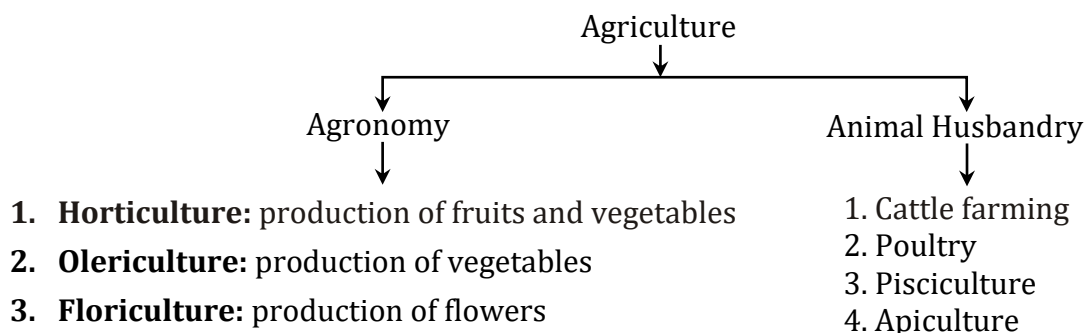
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Different types of crops

Differences between Kharif crops and Rabi crops	
Kharif crops	Rabi crops
These are autumn or monsoon crops. The sowing starts in rainy season.	The sowing starts in winter season.
They are sown in June-July.	They are sown in October-November.
They are harvested in September-October.	They are harvested in March-April.
They require warm and wet weather.	They require dry and cold weather.
They need a lot of water.	They do not need a lot of water.
e.g. Rice, Cotton, Bajra, Jowar, Groundnut, Maize, Spinach, Soyabean, Pigeon pea, Black chana, green gram etc.	e.g. Wheat, Barley, Gram, Pea, Mustard, Linseed etc.

Agriculture: It is an applied biological science which deals with the production of plants and raising of livestock for human use.



The major groups of activities for improving crop yields can be classified as:

- (A) Crop variety improvement (the choice of seed for planting).
- (B) Crop production improvement (the nurturing of the crop plant).
- (C) Crop protection management (the protection of the growing and harvested crops from loss).

A. Crop variety improvement

This approach depends on finding a crop variety that can give a good yield. Varieties or strains of crops can be selected by breeding for various useful characteristics such as disease resistance, response to fertilisers, product quality and high yields.

Methods of crop improvement

- (1) **Hybridisation:** One way of incorporating desirable characters into crop varieties is by hybridisation. Hybridisation refers to crossing between genetically dissimilar plants. The traditional method of crop improvement is by sexual hybridization, making crosses between two genetically dissimilar or diverse plants with desirable characteristics. At fertilization, the genetic information from each parent recombines, so characteristics from each parent are inherited in the offspring. However, many other features are also inherited, along with the desirable ones.

In order to obtain improved variety following steps should be taken.

Choice of parents: Select two varieties of crop plants, each of which has at least one desirable quality, such as high yield or resistance to disease.

Cross breeding: Pollen grains of plants of one variety are dusted over the stigmas of plants of the other variety and vice-versa. It produces a new variety which has good characteristics of both the parents.

Some hybrids/Genetically modified seeds	
Wheat	Kalyan, Sharbati Sonara, Sonalika
Maize	Ganga, Ambar, Jawahar
Rice	Jaya, Padma, Pusa Basmati, IR8
Cotton	G-57, Bt (GM)

Hybridization may be of three types:

- (a) **Intervarietal** (between two different varieties of same species).
- (b) **Interspecific** (between two different species of the same genera).
- (c) **Intergeneric** (between two different genera).

- (2) **Genetic modification or plant transformation:** It is the modern method of crop improvement available to plant breeders. In this, crop is improved by introducing a gene that would provide the desired characteristics.

The plant in which the foreign gene has been introduced is called transgenic plant or genetically modified plant.

For example - Bt cotton is a genetically modified crop which carries bacterial genes that protect the crop plants from insects.



Plant breeding is a branch of science concerned with producing improved varieties of crop plants by introducing several desired characteristics into them.

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The various aspects for which crop variety improvement is carried out are as follows.

- (1) **Higher yield:** To increase the productivity of crop per acre. The main aim of crop variety improvement is to obtain higher yield e.g. grains, tubers. Such a variety is called HYV or high yielding variety.

There is an urgent need to continuously increase crop yield because of

- (i) Rising human population.
 - (ii) Rising animal population.
 - (iii) Increasing realization about intake of proper diet.
 - (iv) Limited cropping area.
- (2) **Better quality:** Quality consideration such as baking quality in wheat, protein quality in pulses, oil quality in oil seeds and preserving quality in fruits and vegetables vary from crop to crop.
 - (3) **Biotic and abiotic resistance:** Crops production can go down due to biotic (microbes, insects, nematodes, rodents) and abiotic stresses (flood, drought, frost, salinity, water logging, heat, cold). Varieties resistant to these stresses can improve crop production.
 - (4) **Change in maturity duration:** The shorter the duration of the crop from sowing to harvesting, the more economical is the variety as
 - (i) It helps the farmers in raising multiple rounds of crops in a year from the same field.
 - (ii) Cost of production is reduced.
 - (iii) Uniform maturity makes the harvesting process easy and reduces losses during harvesting.

- (5) **Wider adaptability:** Developing varieties for wider adaptability will help in stabilizing the crop production under different environmental conditions.
- (6) **Desirable agronomic characteristic:** Developing varieties of desired agronomic characters helps to give higher productivity. To achieve high productivity, the food crops (cereals) should be dwarf (short height), so that they consume less nutrients, become stronger and withstand strong winds (prevent lodging). The fodder crops should be tall and should have profuse branching.

B. Crop production management

India is an agriculture-based country. Farmers have more or less land, money and access to information and technologies –

It is financial conditions that allow farmers to take up different agricultural technologies and farming practices. There is a correlation between yields and higher inputs. So the purchasing capacity for inputs of farmer decides production practices and cropping system.

Therefore, production practices can be at different levels. They include 'no cost' production, 'low cost' production and 'high cost' production practices.

The following are some of the factors that if managed well will help in increasing crop production –

- (1) Nutrient management
- (2) Irrigation
- (3) Cropping patterns

- (1) **Nutrient management:** It is controlling the selection, timing and amount of nutrient supply to the crops. Plant nutrients are inorganic raw materials that the plant absorb from their surroundings and utilize the same in building up their organic matter. Deficiency of these nutrients affects physiological processes in plants including reproduction, growth and susceptibility to diseases. There are many essential elements for growth and development of plants. They are classified into two groups:

Macronutrients: The essential elements utilized by plants in relatively large quantities are called major nutrients or macronutrients.

Micronutrients: The essential elements utilized by plants in relatively small quantities or traces are called micronutrients.

Sources of Nutrients

Sources		Nutrients
1.	Air	Carbon, Oxygen
2.	Water	Hydrogen
3.	Soil	Macronutrients: Nitrogen, Phosphorus, Potassium, Sulphur, Calcium, Magnesium.
4.	Soil	Micronutrients: Iron, Manganese, Boron, Zinc, Copper, Molybdenum, Chlorine.



1. What are the desirable agronomic characteristics for crop-improvement?
2. How do plants get nutrients?
3. Group the following and tabulate them as energy yielding, protein yielding, oil yielding and fodder crop-

Peanut plant, mustard plant, berseem, rice plant.

Mineral replenishment to the soil is done through the addition of manures and fertilizers.

- (i) **Manure:** These are organic substances, obtained from the decomposition of plant and animal wastes.

Advantages of manure

- It increases the number of friendly microbes.
- It improves the texture of soil by adding organic matter (humus). This involves increasing the water holding capacity in sandy soils. In clayey soils, large quantities of organic matter help in drainage and in avoiding water logging.
- It increases soil fertility.
- It reduces soil erosion.
- It is cheap.



Disadvantages of manure

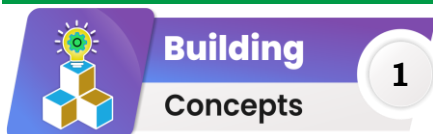
- They have less amount of nutrients as compared to fertilizers.
- Manures are bulky and not easy to store and transport.
- They are not easily absorbed by the plant.
- They are not nutrient specific.
- They are required in large quantity.

Types of manures

Compost: It is a type of manure which is produced from the plant and animal wastes by the action of microbes. The process in which farm waste materials like livestock excreta (cow dung etc.), vegetable waste, animal refuse, domestic waste, sewage waste, straw, eradicated weeds etc, is decomposed in pits is known as composting. The compost is rich in organic matter and nutrients.

Vermicompost: It is the type of manure which is produced by the degradation of organic wastes by using earthworms. This process is known as vermicomposting.

Green manure: It is also a type of organic manure in which some quickly growing legume crops are grown and then mulched by ploughing. They enrich the soil with nitrogen and phosphorus. E.g. Cowpea, Lentil, Sunn Hemp, Dhanicha.



Why are earthworms known as farmer's friend?

Explanation

Earthworms are called farmer's friend because the burrowing action of earthworm helps to loosen the soil particles and then improve the physical structure of soil by adding humus.

(ii) Fertilizers: These are commercially manufactured inorganic salts containing one or more essential plant nutrients like NPK, which are used to increase soil fertility. Fertilizers are a factor in the higher yields of high-cost farming.



Sometimes fertilizers get washed away due to excessive irrigation and are not fully absorbed by the plants. This excess fertilizer then leads to water pollution.

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- ★ Fertilizers should be applied carefully in terms of proper dose, time, and observing pre- and post-application precautions for their complete utilisation.

Advantages of fertilizers

They are nutrient specific and required in small amounts. They are water soluble and absorbed by the plant easily. They are easy to store and transport.



- ★ As we learnt in the previous classes, continuous use of fertilizers in an area can destroy soil fertility because the organic matter in the soil is not replenished and micro-organisms in the soil are harmed by the fertilizers used.

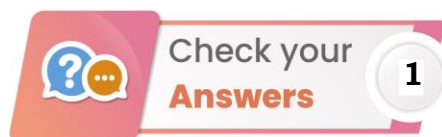
Disadvantages of fertilizers: Fertilizers can change the soil structure by killing the soil microbes. Fertilizers can change the chemical composition of soil. Accumulation of fertilizers in water bodies causes Eutrophication.

Note: Short term benefits of using fertilizers and long-term benefits of using manure for maintaining soil fertility have to be considered while aiming for optimum yields in crop-production.

Organic Farming: It is a kind of farming system in which the harmful chemicals (fertilizers, herbicides and pesticides) are not used. In place of chemicals, the use of organic manures, biofertilizers and biopesticides are recommended. Use of bio-agents such as culture of blue green algae in preparation of biofertilizers, neem leaves or turmeric specifically in grain storage as bio-pesticides, with healthy cropping patterns (mixed cropping, inter-cropping and crop rotation). These cropping systems are beneficial in insect, pest and weed control besides providing nutrients.

Advantages of organic farming:

- | | |
|-------------------|--------------------------------------|
| (i) No pollution | (ii) Healthy and Uncontaminated food |
| (iii) Soil health | (iv) Controlling pests and weeds |



- Tallness and profuse branching are desirable characters for fodder crops.
 - Dwarfness is desired in cereals, so that less nutrients are consumed by plant body.
- Nutrients found in the soil get dissolved in the water from rain and irrigation. This water is absorbed by the roots of plants and transported to different parts of the plants.
- | | | |
|------------------|---|---------------|
| Energy yielding | - | Rice plant |
| Protein yielding | - | Peanut plant |
| Oil yielding | - | Mustard plant |
| Fodder | - | Berseem |



What are alternative methods used by farmers to increase soil fertility?

Explanation

Field fallow: Practice of leaving the field uncultivated for a season.

Crop rotation: Growing of different crops in succession on the same field.

Biofertilizers: Organisms which enrich soil fertility, e.g. *Nostoc*, *Anabaena*, *Rhizobium*.

(2) Irrigation: In India, most agriculture is rain-fed, that is, the success of crops in most areas is dependent on timely monsoons and sufficient rainfall spread through most of the growing season. Hence, poor monsoons cause crop failure. Ensuring that the crops get water at the right stages during their growing season can increase the expected yields of any crop. Therefore, many measures are used to bring more and more agricultural land under irrigation. The process of artificial supplying of water to crop at different intervals is called irrigation. The time and frequency of irrigation varies from crop to crop, soil to soil and season to season.



Why does droughts occur?

Explanation

Droughts occur because of scarcity or irregular distribution of rains. Drought poses a threat to rain-fed farming areas, where farmers do not use irrigation for crop production and depend only on rain. Light soils have less water retention capacity. In areas with light soils, crops get adversely affected by drought conditions. Scientists have developed some crop varieties which can tolerate drought conditions.

Sources of irrigation: Well, tube well, ponds, lakes, river, canal and dams.

Irrigation systems in India

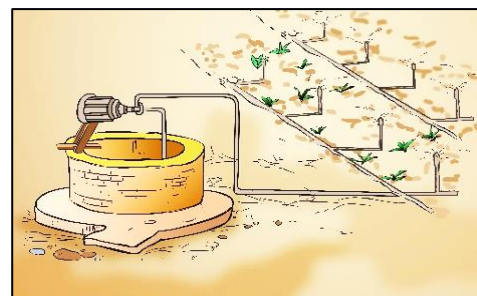
- (i) **Canal system:** This is usually an elaborate and extensive irrigation system. Canals receive water from one or more reservoirs or from the rivers. The main canals are divided into branches which are further divided into distributaries to cover maximum areas for irrigation.
- (ii) **Tanks:** Tanks are small water storage reservoirs which intercept and store the runoff of small catchment area.
- (iii) **Wells**
 - Dug well – Water is collected from water bearing strata.
 - Tube well – Water is pumped up by diesel or electric pumps. They can trap water from the deeper strata.
- (iv) **River valley system:** Due to heavy rainfall, many river valleys are found in Western Ghats and Karnataka. This results in higher run off and discharge flows in the rivers. To prevent this coffee, rubber, coconut are cultivated on the slopes of these valleys and single rice crop is grown at bottom.
- (v) **River lift system:** In areas where canal flow is insufficient or irregular due to inadequate reservoir release, the lift system is more rational. Water is directly drawn from the rivers for supplementing irrigation in areas close to rivers.

Note: Wells and tube wells are successful in those areas where underground water is not saline whereas in those areas where underground water is saline, canal water is used for irrigation.

Modern methods of irrigation: The following modern ways of irrigation are used which help to conserve water.

Sprinkler system

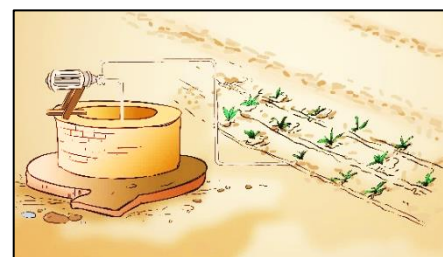
- In this system, the perpendicular pipes having rotating nozzles on top, are joined to main pipeline at regular intervals. When water is allowed to flow through main pipeline under pressure with the help of a pump, it escapes from the rotating nozzle.
- It spreads water uniformly over crop plants and field.
- This method is useful for sandy soil and uneven land, where sufficient water is not available.
- It is an efficient system in the canal irrigated area of Haryana and Rajasthan.
- It is very useful for lawns, coffee plantation and several other crops.



Sprinkler system

Drip System or Trickle system

- It involves the use of pipes fitted with small tubes called emitters. The pipes are laid over or under the soil and emitters release water drop by drop around the roots of the plants.



Drip system

- In this method water is not wasted at all.
- This is the best irrigation technique for fruit crops, garden and trees.
- This method is a boon in poor water regions.

Advantages of irrigation: Supply of essential elements– supplies two essential elements to crops i.e. hydrogen and oxygen. Germination of seeds–Moisture available in soil leads to germination of seeds.

Growth–Growth of roots occur inside the soil by availability of water in soil.

Absorption of nutrients–Water made available to field by the process of irrigation helps in absorption of nutrients by plants from the soil.

Water augmentation: Water availability for irrigation can be assured by augmenting ground water. It is carried out by two methods:

- (i) Rainwater harvesting:** The rain water is collected and used for recharging ground water by sinking deep drain pipes. It can also be poured into wells or used for filling ponds in low level area.
- (ii) Watershed management:** Small check dams are built up in water shed areas to increase percolation of water into ground, reduce flow of rain water and prevent soil erosion.



What are the limitations related to the production of high yielding varieties of crops?

Explanation

High yielding varieties have following disadvantages

1. Production of less fodder
2. High requirement of fertilizers and pesticides
3. Sensitivity to diseases

(3) Cropping patterns

Different ways of growing crops can be used to give maximum benefit.

There are models of raising crops which help in obtaining maximum benefit from the same piece of land, reduce risk of crop failure, disease and infestation. Three common types of cropping patterns are mixed cropping, intercropping and crop rotation.

- (i) Mixed cropping:** It is growing of two or more crops simultaneously on the same piece of land. It is also known as mixed cropping.

Some successful mixed cropping practices are -

Wheat + mustard, wheat + gram, groundnut + sunflower.

Advantages of mixed cropping

- No risk of complete crop failure.
- **Increase in yield:** When cereal or non-leguminous crop is grown along with a leguminous crop, the deficiency of nitrogen in the soil, caused by growing of cereal crop, is compensated by the enrichment of the soil with nitrogen compounds by the growing of leguminous plants. So, it increases the yield.
- **Variety of products:** We can harvest variety of products like pulses, cereals, vegetables or fodder.
- **Improvement in soil fertility:** Growth of leguminous crops improves soil fertility and reduces the requirement of fertilizers.
- **Reduced pest infestation:** The chances of pest infestation are reduced because the pest of one crop rarely finds the same plant nearby.



Mixed farming is the raising of different types of crops and animals on the same farm. It ensures good returns to the farmers and provides easy availability of various articles.

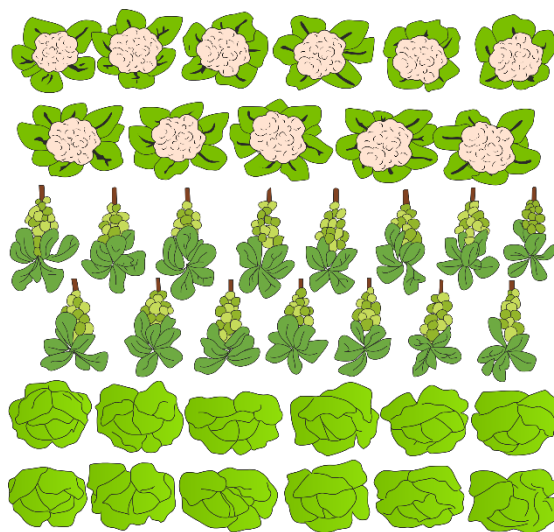
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Disadvantages of mixed cropping

Farmer faces difficulty in applying fertilizer or spraying pesticides on individual crops. Harvesting and threshing of crops separately is not easy, because seeds of the two crops are mixed and then sown (there is no set pattern of rows of crops).

(ii) Inter-cropping: Intercropping is growing two or more crops simultaneously on the same field in a definite pattern. Row patterns like 1: 1, 1: 2 or 1: 3 can be followed. The crops selected for intercropping have different nutrient requirements, different sowing and harvesting dates.

e.g. Soyabean + Maize, Finger Millet (bajra) + Cow Pea (lobia).



Intercropping

Advantages of intercropping

- **Productivity:** Intercropping increases productivity per unit area.
- **Sowing:** Seeds of different crops can be sown separately.
- **No mixing:** There is no mixing of crops.
- **Fertilizers:** Specific fertilizers required for each crop can be added.
- **Pesticides:** Pesticide and weedicide required for each crop can be applied.
- **Harvesting:** Each crop can be harvested, threshed and marketed separately.

Differences between mixed cropping and intercropping

Mixed cropping	Intercropping
Seeds of different crops are mixed before sowing.	Seeds of different crops are not mixed. They are sown separately.
There is no pattern of sowing.	The different crops are sown in separate rows or strips.
It minimizes the risk of crop failure.	It increases crop productivity per unit area.
Only a common type of fertilizer can be added.	Specific fertilizers can be provided to each crop.
Crop specific pesticides cannot be sprayed.	Crop specific pesticides can be sprayed without difficulty.
Harvesting of early maturing crop provides a lot of difficulty.	There is little difficulty in harvesting individual crops.
There is some mixing of the produce of different crops.	There is no mixing of produce of different crops.
Lesser inputs of irrigation and nutrients are required.	Requirement of inputs is comparatively more.

(iii) Crop rotation: The growing of different crops on a piece of land in a preplanned succession is known as crop rotation. The availability of moisture and irrigation facilities decide the choice of the crop to be cultivated after one harvest. If crop rotation is done properly than two or three crops can be grown in a year with good harvests. Depending upon the time duration, crop rotation is done for different crop combination.

A common crop rotation is growing of leguminous plants like peas, pulses, beans etc. alternately with cereal crops like rice, wheat and maize.

Advantages of crop rotation

- Prevents the depletion of soil nutrients.
- Maintains or improves soil fertility.
- Reduces reliance on agricultural chemicals.



1. Which type of hybridization is commonly used in plant breeding?
2. Name one crop which can tolerate water logging in the field.
3. Name three most important mineral elements required for plant growth.

C. Crop protection management

In fields, crops have to be protected from weeds, insects, pests and disease-causing organisms like fungi etc. All these cause damage to crop plants so much that most of the crop is lost. Thus, crops can be protected by the following methods:

- (i) Use of pesticides, insecticides, weedicides
- (ii) Use of resistant varieties
- (iii) Crop rotation and other cropping systems
- (iv) Summer ploughing

- (1) Pest:** Any destructive organism that causes great economic damage or destroy crop plants are called pests.

Examples: Weeds, insects, mites, nematodes etc.



Nematode



How do insect pests attack the crops?

Explanation

The insect pests attack the plants in following ways:

- (i) They cut root, stem and leaves eg. - locusts, caterpillars.
- (ii) They suck the cell sap from various parts of the plant e.g. - aphids.
- (iii) They bore into stem and fruits e.g. - pod borer, stem borer.

Pesticide: It refers to a chemical that is used to kill a pest organism which includes insecticides (for killing the insects), weedicides (for killing the weeds), fungicides (for killing the fungi), nematocides (for killing nematodes) and rodenticides (for killing rodents).



	Pesticide	Affected organism	Example
1	Weedicide	Weeds	2,4-D, Isoproturon
2	Insecticide	Insect	DDT, Malathion
3	Fungicide	Fungal pathogen	Bordeaux mixture
4	Algicide	Algal pathogen	CuSO ₄
5	Rodenticide	Rodents (rat, moles)	ZnSO ₄
6	Nematocide	Nematode	Methyl bromide

Note: 2, 4-D – 2, 4-Dichlorophenoxy acetic acid, DDT-Dichloro-diphenyl-trichloroethane.

Advantages of pesticides: They kill pest quickly, increase food production and are easy to store.

Disadvantages of using pesticides

- (i) They cause soil and water pollution.
- (ii) They destroy even the useful insects.



Be Alert !

★ Residue of pesticides left on the fruits & seeds is harmful for human consumption.

Preventive measures avoiding use of pesticides –

- (i) Use crop rotation, multiple cropping patterns and field fallow.
- (ii) Sowing healthy seeds and summer ploughing.
- (iii) Use of pest and disease resistant hybrid varieties of crop plants.
- (iv) Field also need to be protected against grazing animals by raising wire fences and boundary wall.
- (v) Birds are scared away by raising scarecrows and beating of drums.

Note: Summer ploughing: Ploughing of field all around during hot summer with specialized tools.

It helps to kill weeds, hibernating insects and disease-causing organisms by exposing them to the summer heat.

Scarecrows– It is an image or an effigy in the shape of human being that is placed in field to frighten away birds.

(2) Weed: They are the small-sized unwanted plants which grow along with a cultivated crop in a field. Weeds can severely reduce crop yields by competing for light, space and food. Some common weeds: *Xanthium* (Gokhroo), *Parthenium* (Gajar grass), *Convolvulus*, *Cyprinus rotundus* (Motha), *Amaranthus* (Chaulai), *Chenopodium* (bathua).



Amaranthus



Chenopodium
Common weeds



Convolvulus

Methods of weed control:

Mechanical method: Uprooting, ploughing, burning and flooding.

Biological method: Use of insects or some organisms which consume and destroy the weed plants.

e.g. Prickly-pear cactus (*Opuntia*) is controlled by Cochineal insect and aquatic weeds (*Hydrilla*) are controlled by Carp fish.

Chemical control: Spraying of chemicals like herbicides or weedicides.

E.g. Isoproturon, 2,4-D, Atrazine, MCPA (2-methyl-4-chlorophenoxy acetic acid).



Manual weed control

Crop disease: When the pathogens such as bacteria, fungi and viruses get favourable conditions for growth and propagation, they spread and infect the crop plants causing crop diseases.

Crop diseases are of following types:

(Depending on their mode of occurrence).

Seed borne: Spread through the seed e.g. Ergot of bajra etc.

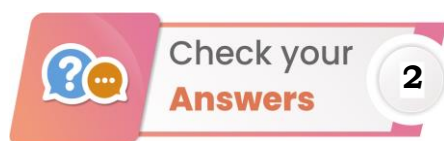
Soil borne: Spread through the soil e.g. Smut of bajra etc.

Air borne: Spread through the air e.g. Rust of wheat etc.

Control of diseases:

The methods of controlling diseases are:

- (i) Seed borne diseases can be prevented by sowing healthy seeds and pre-treating seeds with chemicals to kill pathogens.
- (ii) Soil borne diseases can be prevented by using soil disinfectants and by crop rotation.
- (iii) Air borne diseases can be prevented by spraying specific chemicals, like insecticides, fungicides, antibiotics etc.
- (iv) The breeding of disease resistant crops is a very important method to bring about control of crop diseases.



1. Intervarietal hybridization
2. Rice or Paddy
3. Nitrogen, Phosphorus and Potassium

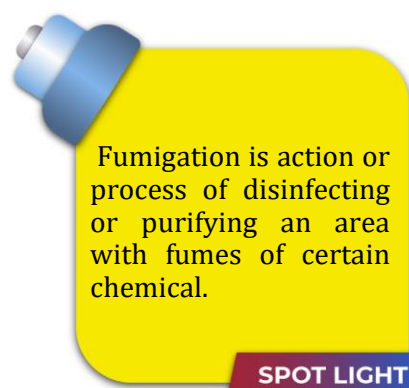
Storage of grains: Storage losses in agricultural produce can be very high. Factors responsible for such losses are -

- (i) **Biotic factors:** e.g. insects, microorganisms (e.g. Fungi, bacteria), mites, birds, rodents and other animals.

- (ii) **Abiotic factors:** e.g. moisture, humidity and temperature.

These factors cause degradation in quality, loss in weight, poor germinability, discolouration of produce, all leading to poor marketability. These factors can be controlled by proper treatment and by systematic management of warehouses.

Preventive and control measures are used before grains are stored for future use. Non-perishable food materials are stored on a commercial scale in gunny bags or in grain silos. They include strict cleaning of the produce before storage, proper drying of the produce first in sunlight and then in shade. Fumigation using chemicals that can kill pests.





Silos for food storage

Animal husbandry: Animal husbandry is the scientific management of animal livestock. It deals with the feeding, caring, disease control and breeding of domesticated animals.

The farming of animals also needs planning to meet their basic requirement to maintain health and control the diseases. However, expanding population and scaling demands have resulted in the limitation in livestock breeding. Therefore, there is a need to improve these techniques which could ensure sustained supply of animals for the rising population without affecting their existence.

Utility of domestic animals		Examples
1.	Milch (Milk-yielding)/Dairy animals	Cow, Buffalo, Goat
2.	Meat & Egg yielding	Goat, Sheep, Fish, Chicken
3.	Draught Animals (used for farm labour)	Horse, Donkey, Mule, Bullock
4.	Hair & Skin yielding	Sheep, Goat, Rabbit, Cow

(1) Cattle farming: Cattle husbandry or cattle farming is carried out primarily for two reasons: milk production by milch animals and for agricultural work, such as ploughing, tilling, irrigation etc, by draught animals. However, they are also domesticated for leather production and meat yield. Indian cattle mainly include cows (*Bos indicus*) and buffaloes (*Bos bubalis*).

Cows are adapted to drier regions of the country, while buffaloes are well adapted to wet areas and river beds. The milk produced by buffaloes has more fat content than that produced by cows. Milk production besides other factors, also depends upon the duration of lactation period, i.e. the period of milk production after the birth of calf.

Therefore, the milk production can be increased by increasing the lactation period.

Breeding in cattle: There are many popular breeds of cows and buffaloes in our country. They can be broadly divided into three categories:

- (i) Indigenous breeds of cows (Desi breeds - they are native to India): e.g. Red Sindhi and Sahiwal (show excellent resistance to diseases).
- (ii) **Foreign or Exotic breeds of cows:** e.g. Jersey, Brown Swiss, Holstein friesian (have a long lactation period).



Sahiwal and Red Sindhi are milch breeds i.e., good milk producing breeds, while Tharparker is a dual purpose breed, i.e. good for both milk and draught.

SPOT LIGHT

- (iii) **Cross breeds of cows:** The breeds chosen for cross breeding have a set of favourable characters that may become a part of the animal that is born as a result of cross breeding, e.g. Karan fries, Frieswal



Cattle Farming

- Some indigenous breeds of buffaloes are Murrah, Surti, Mehsana and Nagpuri.
- Dr. V. Kurien was the founder chairman of the National Dairy Development Board (NDDB) Gujarat from 1965 to 1998. He started the "Operation flood" and is also known as father of white revolution in India.

Farm Management Practices: Farm Management Practices are aimed at farming of animals, looking after their health as well as production of clean milk. It has four components.

(A) Grooming (B) Shelter (C) Feeding (D) Health care

(A) Grooming of cattle

- Cattles must be regularly groomed.
- Removing dirt, loose hair and external parasite.

(B) Shelter for cattle

- Providing suitable shelter to cattle is an important part of cattle husbandry.
- Cattle shelters are well demarcated, partially or completely covered areas where animals can live comfortably and are protected from predators as well as natural factors (heat, cold, rain etc.)

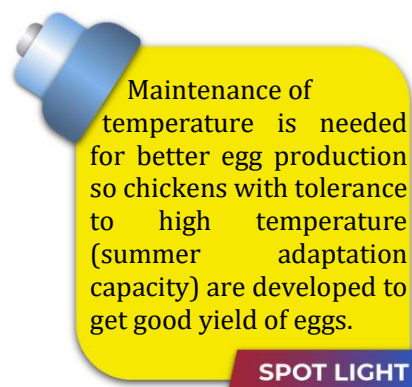
Characteristics of good animal shelter

- It should be spacious.
- The shelter should be clean, dry, airy and well ventilated (The floor of the cattle shed needs to be sloping as to stay dry and to facilitate cleaning).
- There should be proper arrangement for the hygienic disposal of animal waste.
- It should have arrangement of fresh, clean drinking water.

(C) Feeding of cattle

The "food" that contains essential components needed for the growth, development and general maintenance of body of domesticated animal is called feed.

The feed requirement of an animal depends upon age, growth, pregnancy, lactation, nature of work and state of health.



Feeding management: The cattle require two kinds of food.

(i) Maintenance requirement: It contains all the nutrients for the maintenance of health of cattle. It enables the animals to perform all the basic functions of life, provides strength and makes them resistant to diseases.

(ii) Milk producing requirement: It contain the specific nutrients required during lactation period. It thus increases the duration of lactation period and increases the milk yield. These food requirements are fulfilled with a specific diet containing the following components.

Roughage: Fibre- rich food, such as hay, green fodder, silage and legumes, e.g. cowpea, berseem.

Concentrates: Nutrient rich food but low in fibres. It contains carbohydrates, vitamins and especially high amount of proteins. e.g. Maize, oats, barley, jowar, gram and their by products such as wheat bran, rice bran, gram husk, oil seed cakes, molasses.

Feed additives: These are micronutrients, hormones etc, that promote health and milk output of dairy animals.

(D) Disease management: A healthy animal feeds regularly and has a normal posture. For prevention against viral and bacterial diseases vaccinations must be given to farm animals.

Parasites of Cattle: External parasites - live on skin and cause skin diseases.

Internal parasites - e. g. worms and flukes. Worms affect stomach and intestine, while flukes infect and damage the liver.

Infectious Diseases: They can spread to other animals on the farm and, can sometimes infect humans too. These diseases can be bacterial (Anthrax, Tuberculosis) or viral (Cow pox, Foot and mouth disease).



Which method is commonly used for improving the cattle breeds and why?

Explanation

Cross breeding indigenous breeds with exotic breeds. Foreign or exotic breeds have higher milk yield and longer lactation period as compared to indigenous breeds whereas the local breeds are hardy and resistant to several diseases. Therefore, indigenous breeds should be cross-bred with exotic breeds.

(2) Poultry farming: Poultry farming is undertaken to raise domestic fowls for egg production and chicken meat. They are supposed to be the best sources of proteins and fats. Along with these birds, now a days, other birds such as ducks, geese, pigeons, turkeys etc., are also reared for increased production of eggs and meat. The birds reared for egg production are called eggers or layers while those for meat purpose are known as broilers.

Breeds of hen Indigenous breeds (Desi breeds - they are native to India): e.g. Aseel, Busra (show excellent resistance to diseases)

Foreign or Exotic Breeds: (lay more number of eggs than indigenous breeds and also give more meat) e.g. White Leghorn, Rhode Island Red.

Cross breeds: The desirable traits which can be achieved by cross breeding are as follows –

- (i) Increased fertility and fecundity (egg laying capacity) of the fowl.
- (ii) Large sized eggs.
- (iii) Increased meat yield.
- (iv) Low maintenance requirements.
- (v) Less diet requirements.
- (vi) Rise in number and quality of chicks.
- (vii) Increase in tolerance to high temperature or summer adaptation capacity.
- (viii) Reduced size of layers with ability to utilize more fibrous and cheaper diets made by using agricultural by products.
- (ix) Dwarf broiler parent for commercial chick production.

Care of layers and broilers:

Layers and broilers require special care for nutrition, shelter and disease control. The management and care differs in different phases of their lives.

Layers: Egg production is the most efficient economic aspect of poultry. Hens (female fowls) raised for egg production are called layers.

During growing period, they require sufficient space to avoid overcrowding enabling proper growth. During egg laying period, they need enough space and light which increases the fertility.



Aseel



Leghorn

Broilers

Broilers are quick growing table birds which are raised for about 7-8 weeks to attain proper weight. When the chickens are grown as broilers, their housing, nutritional and environmental needs become different from that of the layers. At this time, they are provided with the conditions which are favourable for their rapid growth & low mortality and to maintain feathering and carcass quality. The food provided to them should be protein rich (55 - 70%) with adequate amount of fats and high amount of vitamin A and K.



What would happen if layers are larger in size and no summer adaptation capacity? In order to get small sized poultry birds having summer adaptability, what method will be employed?

Explanation

Large sized birds require more feed. No summer adaptation is connected with reduced egg laying capacity. In order to get small sized poultry birds having summer adaptability, it is desirable to -

- (i) Either introduce the required exotic birds from outside.
- (ii) Cross breed the local birds with exotic birds from outside.

Small sized birds are preferred for

- (i) Low requirement of feed.
- (ii) High egg laying capacity.
- (iii) Low requirement for space.

Poultry care: To produce good quality birds, good management practices are required which are different for broilers and layers. These include:

- (i) Poultry feed.
- (ii) Hygienic condition in housing.
- (iii) Maintenance of temperature.
- (iv) Prevention and control of disease, as well as pests.
- The feed given to poultry birds consists of mashed cereals like bajra, maize, wheat, jowar, ragi, rice bran and groundnut cakes. The 'fish meal' (prepared from the wastes of meat processing industry) is also used to feed poultry birds.

Disease management: Poultry birds suffer from a number of viral, fungal and bacterial diseases. The birds may also be affected by parasites and nutritional deficiencies. The birds can be protected from various diseases by adopting the following measures:

- (i) Proper cleaning and sanitation of their living area.
- (ii) Spraying of disinfectants at regular intervals.
- (iii) Shelter should be cleaned properly and regularly for prevention against viral and bacterial diseases.
- (iv) Vaccinations must be given to these birds.
- (v) Every animal should be vaccinated to immunize it against common infectious diseases.



1. What management practices are common in dairy and poultry farming?
2. What are the differences between broilers and layers and in their management?



Discuss the implications of the following statement "It is interesting to note that poultry is India's most efficient converter of low fibre food stuff (which is unfit for human consumption) into highly nutritious animal protein food."

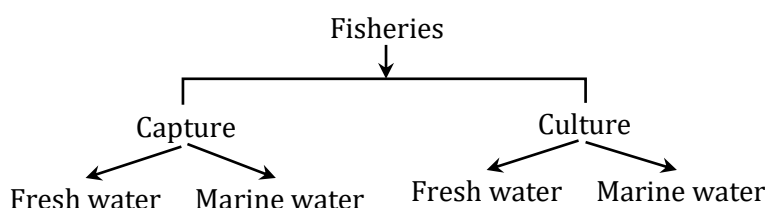
Explanation

India is basically an agricultural society where a lot of wastes are produced during food processing, e.g., fish meal, meat meal, rice bran, etc. They are profitably used in forming poultry feed. In return poultry provides us with egg and meat rich in animal protein.

(3) Fish production:

Fisheries are establishments connected with capture, prevention, exploitation and utilization of various types of fish, prawn, lobsters, oysters etc. Fish is a cheaper source of animal protein of our food.

Depending upon the mode of obtaining fish, fisheries are of two types



- (i) **Capture fishery:** The fish is caught from natural water, both marine and inland. Modern technology is used in capturing of fish.
- (ii) **Culture Fishery:** It is the culture of aquatic organisms with controlled environment and scientific knowledge for commercial production.

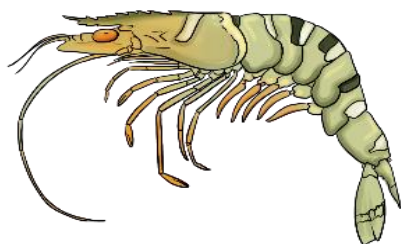
Marine Fisheries: They are fish catching areas in sea. India has an extensive coastline (about 7500 km) and an exclusive sea zone beyond it. Popular marine fishes are Pomphret, Mackerel, Tuna, Sardines and Bombay duck. These fishes are captured from coastal areas. Marine fishes are caught using fishing nets from boats and large trawlers. Yields are further increased by locating large schools of fish using satellites and eco-sounders. Mechanized fishing boats are also introduced for fishing.



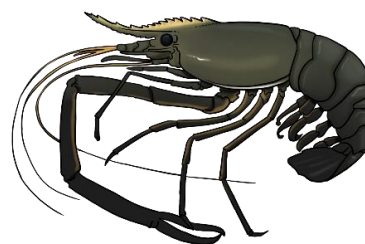
Fish culture

Mariculture: To fulfill the demands of growing population and to sustain the number of fish in marine waters, mariculture is practiced. Some marine organisms of high economic value are also farmed in sea water. This includes mullets, *bhetki*, pearl spots, shellfish such as prawns, mussels, oysters and even seaweed. Oysters not only provide food but pearls too.

Mariculture is growing rapidly as the demand for fish is increasing while the stock is declining due to excessive exploitation from sea.



Macrobrachium rosenbergii (Fresh water)



Penaeus monodon (Marine)

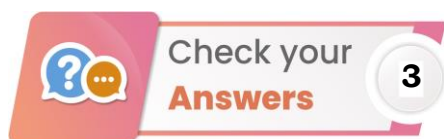
Inland Fisheries (Fresh Water)

India has a large number of inland water resources, e.g. rivers, ponds and lakes. The mangrove wetlands, brackish water areas (where sea water and fresh water mix together) in lagoons and estuaries also constitute large inland water reservoirs.

Aquaculture

Capture fisheries are poor in inland waters, Therefore, major fish production from inland water is carried out through culture called aquaculture. Aquaculture is the scientific rearing and management of aquatic animals and plants in both fresh and marine water.

Fish culture is sometimes carried out in paddy fields. These fish are grown in shallow waters of the paddy fields and can tolerate high temperature and turbidity.



1. (i) Proper shelter, its hygiene, aeration and lighting.
(ii) Proper feed and feed additives, clean drinking water.
(iii) Health care including vaccination.

2. Difference between broilers and layers:

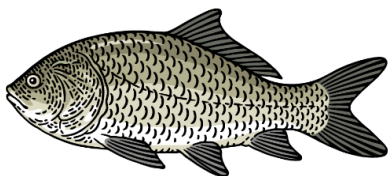
Broilers		Layers	
(1)	Fast growing young chicken 6-10 weeks age which are known for the good quality and taste of their meat.	(1)	Layers are mature hens raised for egg laying.
(2)	Broilers are given diet rich in protein, vitamin A and K, with adequate amount of fat.	(2)	Layers do not require specific nutrients in diet.
(3)	They need space, proper hygiene and temperature for optimum growth.	(3)	Light is required for good egg laying, temperature variations may occur to some degree.

Composite fish culture: Composite fish culture is an extensive fish farming system, where fishes of different species differing in feeding habit and habitats are stocked and cultured in the same pond. It is known as polyculture and mixed fish farming. These cultures leads to the efficient utilization of food present in the pond and thus, increase the yield. Both local and imported fishes are used in such cultures.

In composite fish culture, 5-6 species of fishes are used. Out of these, three are exotic species, including silver carp, grass carp and common carp. While other three are Indian species Catla, Rohu and Mrigal.

The advantage of composite fishes culture over single species culture are as follows

- (i) All available microhabitats (living areas) are fully utilized, as each species occupy different habitat.
- (ii) The different species of fishes do not harm each other. Instead, they can have beneficial effects on each other.
- (iii) Different species have different food habits and thus, do not compete for food.
- (iv) Composite culture yield variety of fishes and that, to more in number as compared to the single species culture.



Catla (Surface Feeder)



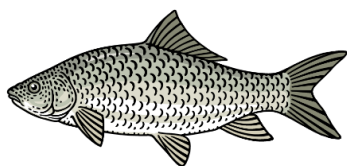
Silver Carp (Surface Feeder
Feeds on phytoplanktons)



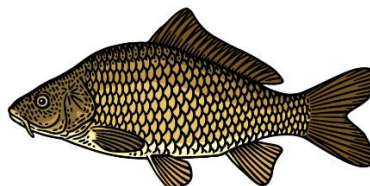
Rohu (Column Feeder/
middle zone)



Grass Carp (Surface, column
& bottom feeder)



Mrigal (Bottom Feeder)



Common Carp (Bottom Feeder)

When all the above species are stocked together in a pond, each one occupies its own niche (living area) and feeds upon food different from each other. It increases the yield of all species of fishes. One problem associated with such composite culture is that many of these fishes breed only during monsoon season. Therefore, good quality seeds (fish seed is fish egg) of these fishes may not be available or get mixed with that of other species, if collected from wild. This problem can be solved by breeding of these fishes in ponds by hormonal stimulation.

This technique supplies pure and good quality fish seeds in desirable quantities for the culture.



1. What are the advantages of composite fish culture?
2. What are the desirable characters of bee varieties suitable for bee keeping?
3. How are fishes obtained?

(4) Bee keeping

Bee keeping is the rearing, care and management of honey bees for obtaining honey, wax and other substances.

Honey is an economically important product obtained from the honeybees. It is widely used hence, bee-keeping is practiced at a large scale. Bee-keeping requires very low investments making it an additional income generation activity. Further, this practice is a source of beeswax which is used in different medicines and cosmetics. Bee-keeping is one of the oldest agricultural pursuits of man. Honey is mentioned as a medicine in ancient Indian literature. Now, honeybees have reached the highest degree of domestication. The modern scientific methods for the production of honey and beeswax is known as apiculture. The honeybees used for apiculture are

Common species of honey bee

1. *Apis cerana Indica* (Indian bee)
2. *Apis dorsata* (Grant bee or rock bee)
3. *Apis florea* (Little bee)
4. *Apis mellifera* (European and Italian bee)

Apis mellifera is more preferred bee as it produces more honey per hive per year.

This is the variety commonly used for commercial honey production. They also breed very well, in the hive for a long duration and sting less.

The Italian bees have high honey collection capacity. They sting somewhat less. They stay in a given beehive for long periods and breed very well. Honey bees are reared in artificial hives/ bee farms i.e. apiary.

This hive is made of wood and has large chamber. This artificial and movable hive is known as apiary. The chambers are partitioned with wax sheets to attract bees. During extraction of honey, comb sheets are taken out and kept in honey extractor. It throws out honey by centrifugal force without damaging the comb yielding large amount of honey.

The value or quality of honey depends upon the pasturage, or the flowers available to the bees for nectar and pollen collection. In addition to adequate quantity of pasturage, the kind of flowers available will determine the taste of the honey.



Apiary

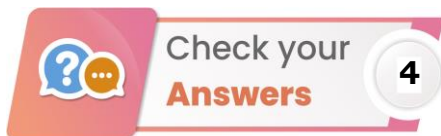


Honey extractor

Pasturage

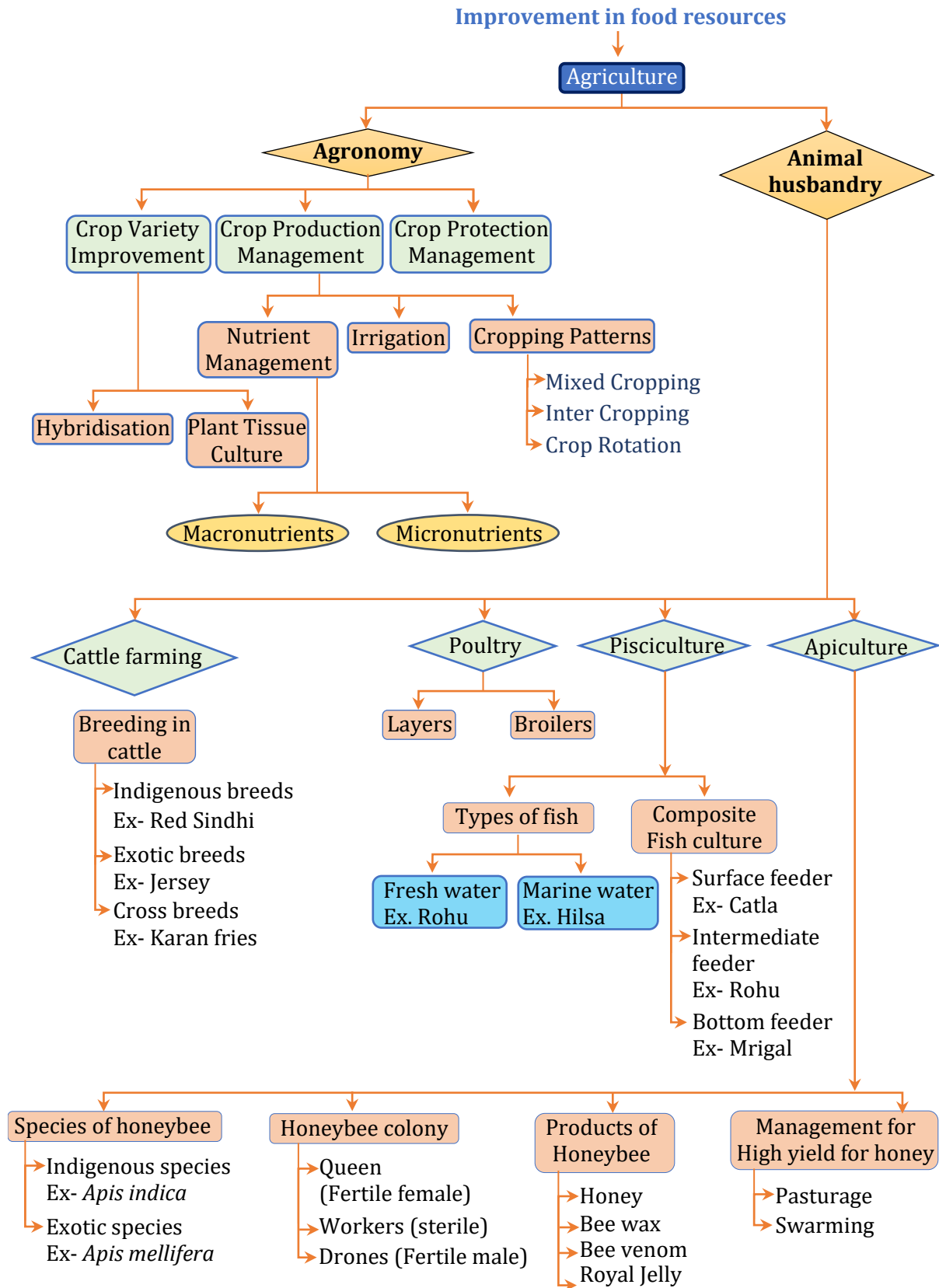
It is the vegetation cover in a particular area. It determines the type and number of flowers available to the bees for collection of nectar and pollen.

- Apiary is an artificial and movable beehive.
- Honey bees produce beeswax from eight paired glands on the underside of their abdomen.
- Three types of honey bees in a hive are
 1. Queen - Fertile female which lays eggs.
 2. Drones - Males which mate with queen.
 3. Workers - They are sterile females and are the smallest member of the family. They collect nectar and pollens from the flower.



1.
 - (i) There is no competition for food or space amongst different types of fish.
 - (ii) Food available in different part of the pond are utilized due to their different food habits and different habitats.
 - (iii) The fish yield is high as 5-6 types of fish are growing simultaneously.
2.
 - (i) Gentleness in nature
 - (ii) Good honey collection
 - (iii) Prolific queen
 - (iv) Less swarming
 - (v) Ability to protect itself from enemies.
- 3.. There are two methods of obtaining fish, capture fishery (capturing fish) from natural waters and culture fishery in impounded waters. In both cases the fish are caught with the help of nets etc.

Chapter At a Glance



SOME BASIC TERMS

1. **Hybridization** – Refers to crossing between genetically different or dissimilar plants.
 2. **Intergeneric** – Occurring between two or more genera.
 3. **Soil Salinity** – A condition in which concentration of soluble salt in the soil becomes high.
 4. **Bio-agents** – Organisms used in agriculture to control pests, diseases and weeds.
 5. **Germinability** – Tendency of a seed to germinate.
 6. **Milch Animal** – Milk producing female cattle.
 7. **Draught Animal** – The animal used for farm labour.
 8. **Exotic breeds** – Animals which are imported from foreign land.
 9. **Photoperiod** – The period of time in a day that in an organism exposed to light.
 10. **Vaccination** – Administration of a vaccine to help in developing the immunity against a disease.
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